

tion of an intelligent algorithm that can analyse the number of objects and can correct the shadow effect. It can achieve a high angular resolution of 0.042° at a scan rate of up to 50Hz. The lidar scanner detects the contour of packages on the conveyor belt and uses that information to calculate the conveyor belt utilisation. The product is easy to install and commission, making the Contour2D a fast and simple solution to employ.

Pepperl + Fuchs
<https://pepperl-fuchs.com>

DisplayPort KVM over IP Extender

ATEN Advanced KX9970 is a 5K DisplayPort KVM over IP Extender. The KX9970 transmitter and receiver is capable of enabling real-time KVM over IP with resolutions up to 5120 × 1440@60 (4:4:4) and 5120 × 2880@30 (4:4:4) lossless video compression over long distances. It provides a zero-latency transmission and colour depth up to 12 bits. The extenders also provide USB isochronous transfer that enables USB camera and USB speaker usage between transmitter and receiver. The KX9970 is suitable for various applications, which include diverse control room applications such as air-traffic control, eSports applications, and more.

Aten
www.aten.com

Cellular-to-cloud IoT kits

CK-RA6M5 and CK-RX65N from Renesas are two new cellular-to-cloud IoT development platforms. The kits are powered by RA and RX families of 32-bit microcontrollers (MCUs) and feature a RYZ014A Cat-M1 Pmod module—a certified LTE cellular module that offers the ability to establish a wireless connection between MCUs and cloud services quickly and securely without a gateway. These kits incorporate multiple sensors, hardware based



security, and a reliable software stack. In addition, the cloud kits can utilise the global cloud service platforms such as AWS Cloud and other IoT services. Both kits are designed to run on Amazon Web Services, FreeRTOS for CK-RA6M5 using flexible software package, and CK-RX65N using RX driver package.

Renesas
<https://www.renesas.com>

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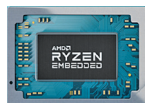
OS-independent hypervisor

Real-Time Systems (RTS) has announced RTS Safe Hypervisor, an OS-independent functional safety certified Type 1 hypervisor to target mixed-critical workloads based on x86 multicore processor technologies. RTS Safe Hypervisor is planned to be delivered as a complete OEM package, bundling the certified real-time hypervisor with functionally safe and non-safe virtual machines and a certified safe OS, such as the Linux based Zephyr or QNX. The Hypervisor allows users to combine any operating system with any available processor from Intel. It enables you to virtualise devices and their IoT connectivity individually.

RTS
<https://www.real-time-systems.com>

New Embedded R2000 SoC

AMD 2nd generation of Ryzen Embedded R2000 processor is specially designed for embedded and industrial platforms. The mid-range system-on-chip (SoC) is optimised for a wide range of applications, such as ML, IoT, industrial, and robotics systems. It is capable of powering up to four 4K displays simultaneously and supports OS from both Microsoft and Linux. The module is scalable up to four Zen+ CPU cores with eight threads, 2MB of L2 cache, and 4MB of shared L3 cache. The R2000 series processor supports up to



3200MT/s DDR4 dual-channel memory and features enterprise-class security features, which include AMD Secure Processor to help protect sensitive data and validate code before it is executed and offers AMD Memory Guard for real-time DRAM memory encryption.

AMD
<https://www.amd.com>

MISCELLANEOUS

Halogen-free flux

SMEF-Z52 flux from SHENMAO America, Inc. is an enhanced solder-joint encapsulation material, which provides higher bonding strength and joint protection. The SJEM flux combines the ability of conventional flux and underfill, thus providing greater joint protection along with an excellent bonding strength as the epoxy cures after the reflow process. This flux can be used for a wide variety of soldering processes, which include jetting, dipping, stencil printing, dispensing, and pin-transfer processes. It is suitable for multiple IC packages, such as wafer-level-package, system-in-package, and flip-chip. The halogen-free (RELO) flux complies with RoHS, RoHS 2.0, and REACH.

SHENMAO America, Inc.
www.shenmao.com

Syringe with NFC tag

NP Plastibell has revealed a truly innovative syringe with near-field communication (NFC) tag embedded in it. The NFC tag used in this syringe is from STMicroelectronics, which is directly moulded into the syringe. The tag enables doctors, patients, and staff to access important medicine-related information stored in it. The syringe is safe for medical use as it is sterilised at a temperature above 130°C by autoclave without damaging the electronics.



NP Plastibell
<https://www.clayens-np.com>